

大数据处理综合实验课程

实验四：MapReduce 高级编程技术

姓名 杨力泉

学号 231220069

日期 2026 年 5 月 14 日

一、本地实验环境

- 操作系统：Ubuntu 22.04 (WSL2)
- JDK 版本：OpenJDK 1.8.0_281
- Hadoop 版本：3.2.1
- 硬件配置：vCPU2核，内存4GB，硬盘30G
- 集群环境：课程实验平台

二、任务一：统计喜欢相同电影的用户

2.1 设计思路

目标：找出所有共同喜欢电影的用户对，并统计他们共同喜欢的电影数量。
输入：ratings.csv (userId, movieId, rating, ...) ， rating ≥ 4.0 视为喜欢。
输出：A-B N，其中 A < B，N 为共同喜欢电影数。

采用两个串联 MapReduce 作业实现：

- **Job1：**生成每对喜欢同一部电影的用户对，每个用户对输出计数值 1。
 - **Mapper：**过滤 rating ≥ 4.0 的记录，输出 <movieId, userId>。
 - **Reducer：**对同一 movieId 的用户列表排序，生成所有无序对 A-B，输出 <A-B, 1>。
- **Job2：**将 Job1 的临时结果按用户对聚合求和。
 - **Mapper：**读取 <A-B, 1>，恒等输出。
 - **Reducer：**对相同的用户对累加计数，得到最终 N。

Key/Value 类型：

作业	阶段	Key	Value
Job1	Map	Text(movieId)	Text(userId)
Job1	Reduce	Text(A-B)	IntWritable(1)
Job2	Map	Text(A-B)	IntWritable
Job2	Reduce	Text(A-B)	IntWritable(N)

2.2 核心伪代码

```
// Job1 Mapper
map(key, value):
    fields = value.split(",")
    if fields[2] >= 4.0:
        emit(fields[1], fields[0])

// Job1 Reducer
reduce(movieId, users):
    sort(users)
    for i in 0..len-1:
        for j in i+1..len-1:
            pair = users[i] + "-" + users[j]
            emit(pair, 1)

// Job2 Reducer
reduce(pair, counts):
    sum = 0
    for c in counts: sum += c
    emit(pair, sum)
```

2.3 作业提交与执行

```
yarn jar Task1.jar Task1 \
    /user/root/Exp4/ratings.csv \
    /user/231220069a/Exp4/task1_output
```

集群提交过程截图：

1	231220069a	yarn jar /home/231220069a/Exp4/Task1....	MR	2026/5/14 11:30:00	2026/5/14 11:30:46	 成功	查看日志					
application_1774860907544_2829	231220069a	Task1 Step2 Sum	MAPREDUCE	2025class02 0	Thu May 14 11:30:25 +0800 2026	Thu May 14 11:30:25 +0800 2026	Thu May 14 11:30:44 +0800 2026	FINISHED	SUCCEEDED	N/A	N/A	N
application_1774860907544_2828	231220069a	Task1 Step1 Generate Pairs	MAPREDUCE	2025class02 0	Thu May 14 11:30:05 +0800 2026	Thu May 14 11:30:05 +0800 2026	Thu May 14 11:30:20 +0800 2026	FINISHED	SUCCEEDED	N/A	N/A	N

YARN WebUI 执行报告（两个 Job 均成功）：

Tracking URL: History					
Log Aggregation Status: SUCCEEDED					
Application Timeout (Remaining Time): Unlimited					
Diagnostics:					
Unmanaged Application: false					
Application Node Label expression: <Not set>					
AM container Node Label expression: <DEFAULT_PARTITION>					
Application Metrics					
Total Resource Preempted: <memory:0, vCores:0>					
Total Number of Non-AM Containers Preempted: 0					
Total Number of AM Containers Preempted: 0					
Resource Preempted from Current Attempt: <memory:0, vCores:0>					
Number of Non-AM Containers Preempted from Current Attempt: 0					
Aggregate Resource Allocation: 53174 MB-seconds, 29 vcore-seconds, 0 yarn.io/gpu-seconds					
Aggregate Preempted Resource Allocation: 0 MB-seconds, 0 vcore-seconds					
Show 20 entries Search:					
Attempt ID	Started	Node	Logs	Nodes blacklisted by the app	Nodes blacklisted by the system
appattempt_1774860907544_2828_000001	Thu May 14 11:30:05 +0800 2026	http://slave222:8042	Logs	0	0
Tracking URL: History					
Log Aggregation Status: SUCCEEDED					
Application Timeout (Remaining Time): Unlimited					
Diagnostics:					
Unmanaged Application: false					
Application Node Label expression: <Not set>					
AM container Node Label expression: <DEFAULT_PARTITION>					
Application Metrics					
Total Resource Preempted: <memory:0, vCores:0>					
Total Number of Non-AM Containers Preempted: 0					
Total Number of AM Containers Preempted: 0					
Resource Preempted from Current Attempt: <memory:0, vCores:0>					
Number of Non-AM Containers Preempted from Current Attempt: 0					
Aggregate Resource Allocation: 63630 MB-seconds, 36 vcore-seconds, 0 yarn.io/gpu-seconds					
Aggregate Preempted Resource Allocation: 0 MB-seconds, 0 vcore-seconds					
Show 20 entries Search:					
Attempt ID	Started	Node	Logs	Nodes blacklisted by the app	Nodes blacklisted by the system
appattempt_1774860907544_2829_000001	Thu May 14 11:30:25 +0800 2026	http://slave207:8042	Logs	0	0

2.4 输出结果

HDFS 输出路径：[/user/231220069a/Exp4/task1_output](#)

部分结果截图：

```
1-101
1-100 17
1-101 12
1-102 8
1-103 41
1-104 12
1-105 26
1-106 1
1-107 4
1-108 5
1-109 8
1-1111
1-110 10
1-111 8
1-112 13
```

取消

下载

三、任务二：有相同喜好的用户

3.1 设计思路

目标：基于任务一的输出，生成每个用户的同好列表，格式 `A:B,C,D`，好友 ID 按数值升序排列。

输入：任务一输出 `A-B \t N`。

输出：`A:B,C,D...`

采用一个 **MapReduce** 作业完成：

- **Mapper：**解析 `A-B \t N`，双向输出 `<A, B>` 和 `<B, A>`，使用 `IntWritable` 作为 Key，保证 Shuffle 按用户 ID 数值排序。
- **Reducer：**对同一用户的所有好友使用 `TreeSet<Integer>` 去重并数值排序，最后拼接为 `用户ID:好友1, 好友2, ...` 输出。
- 设置 `setNumReduceTasks(1)`，使输出文件内部整体有序。

Key/Value 类型：

阶段	Key	Value
Map	IntWritable(用户)	Text(好友)
Reduce	输出: Text(整行)	Text(空)

3.2 核心伪代码

```
// Mapper
map(key, line):
    parts = line.split("\t")
    users = parts[0].split("-")
    if users.length == 2:
```

```
emit(IntWritable(users[0]), Text(users[1]))
emit(IntWritable(users[1]), Text(users[0]))

// Reducer
reduce(userId, friends):
    TreeSet<Integer> set
    for f in friends: set.add(Integer.parseInt(f))
    out = userId + ":" + join(set, ",")
    emit(out, "")
```

3.3 作业提交与执行

提交命令

```
yarn jar Task2FromTask1.jar Task2FromTask1 \
/user/231220069a/Exp4/task1_output \
/user/231220069a/Exp4/task2_output
```

集群提交过程截图：

1	231220069a	yarn jar /home/231220069a/Task2FromTask1.jar Tas...	MR	2026/5/14 19:20:04	2026/5/14 19:20:25	成功	查看日志							
application_1774860907544_2847	231220069a	Task2 from Task1 output (sorted)	MAPREDUCE	2025class02	0	Thu May 14 19:20:07 +0800	Thu May 14 19:20:07 +0800 2026	Thu May 14 19:20:23 +0800	FINISHED	SUCCEEDED	N/A	N/A	N/A	N/

YARN WebUI 执行报告：

Queue: [/user/231220069a](#)

FinalStatus Reported by AM: SUCCEEDED

Started: Thu May 14 07:20:07 -0400 2026

Launched: Thu May 14 07:20:07 -0400 2026

Finished: Thu May 14 07:20:23 -0400 2026

Elapsed: 16sec

Tracking URL: [History](#)

Log Aggregation Status: SUCCEEDED

Application Timeout (Remaining Time): Unlimited

Diagnostics:

Unmanaged Application: false

Application Node Label expression: <Not set>

AM container Node Label expression: <DEFAULT_PARTITION>

Application Metrics

Total Resource Preempted:	<memory:0, vCores:0>
Total Number of Non-AM Containers Preempted:	0
Total Number of AM Containers Preempted:	0
Resource Preempted from Current Attempt:	<memory:0, vCores:0>
Number of Non-AM Containers Preempted from Current Attempt:	0
Aggregate Resource Allocation:	54598 MB-seconds, 30 vcore-seconds, 0 yarn.io/gpu-seconds
Aggregate Preempted Resource Allocation:	0 MB-seconds, 0 vcore-seconds

Show 20 entries

Attempt IDStartedNodeLogsNodes blacklisted by the appNodes blacklisted by the system

appattempt_1774860907544_2847_000001	Thu May 14 19:20:07 +0800 2026	http://slave208:8042	Logs	0	0
--------------------------------------	--------------------------------	----------------------	------	---	---

3.4 输出结果

HDFS 输出路径：`/user/231220069a/Exp4/task2_output`

部分结果截图：

1:2,3,4,5,6,7,8,9,10,11,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120,121,122,123,124,125,126,127,128,129,130,131,132,133,134,135,136,137,139,140,141,142,143,144,145,147,148,149,150,151,152,154,155,156,157,158,159,160,161,162,163,164,165,166,167,168,169,170,171,172,173,174,176,177,178,179,180,181,182,183,185,186,187,188,189,190,191,192,193,194,195,196,197,198,199,200,201,202,203,204,205,206,207,208,209,210,211,212,213,215,216,217,218,219,220,221,222,223,224,225,226,227,228,229,230,231,232,233,234,235,236,237,238,239,240,241,242,243,244,246,247,248,249,250,251,252,253,254,255,256,258,260,261,262,263,264,265,266,267,268,269,270,271,272,273,274,275,276,277,278,279,280,281,282,283,284,285,286,287,288,289,290,291,292,293,29

取消

下载

四、任务三：统计电影热度

4.1 设计思路

目标：统计 `top100.csv` 中 100 部电影各自有多少人喜欢（rating ≥ 4.0 ）。

输入：`ratings.csv` 和 `top100.csv`（通过 DistributedCache 分发）。

输出：电影ID:喜欢人数。

采用一个 MapReduce 作业：

- **Mapper.setup()**：从 DistributedCache 读取 `top100.csv`，将 100 个电影 ID 存入 HashSet。
- **Mapper.map()**：过滤 rating ≥ 4.0 且电影 ID 在 top100 集合中的记录，输出 `<movieId, 1>`。
- **Reducer**：对每个电影 ID 的计数求和，输出 电影ID:总人数。

Key/Value 类型：

阶段	Key	Value
Map	Text(movieId)	IntWritable(1)
Reduce	输出: Text(F:N)	Text(空)

4.2 核心伪代码

```
// setup()
for cacheFile in getLocalCacheFiles():
    if "top100.csv":
        read lines → topSet.add(movieId)

// map()
if rating >= 4.0 and movieId in topSet:
```

```
emit(movieId, 1)

// reduce()
sum = 0
for v in values: sum += v
emit(movieId + ":" + sum, "")
```

4.3 作业提交与执行

提交命令（通过 `-files` 将 `top100.csv` 加入缓存）：

```
yarn jar Task3.jar Task3 \
  -files hdfs:///user/root/Exp4/top100.csv#top100.csv \
  /user/root/Exp4/ratings.csv \
  /user/231220069a/Exp4/task3_output
```

集群提交过程截图：

1	231220069a	yarn jar Task3.jar Task3 -files hdfs:///user/root/Exp4/...	MR	2026/5/14 19:43:33	2026/5/14 19:43:55	成功	查看日志						
application_1774860907544_2848	231220069a	Task3 Movie Hotness	MAPREDUCE	2025class02	0	Thu May 14 19:43:37 +0800 2026	Thu May 14 19:43:52 +0800 2026	FINISHED	SUCCEEDED	N/A	N/A	N/A	N/A

YARN WebUI 执行报告：

Application Overview

User: 231220069a

Name: Task3 Movie Hotness

Application Type: MAPREDUCE

Application Tags:

Application Priority: 0 (Higher Integer value indicates higher priority)

YarnApplicationState: FINISHED

Queue: 2025class02

FinalStatus Reported by AM: SUCCEEDED

Started: Thu May 14 07:43:37 -0400 2026

Launched: Thu May 14 07:43:37 -0400 2026

Finished: Thu May 14 07:43:52 -0400 2026

Elapsed: 14sec

Tracking URL: History

Log Aggregation Status: SUCCEEDED

Application Timeout (Remaining Time): Unlimited

Diagnostics:

Unmanaged Application: false

Application Node Label expression: <Not set>

AM container Node Label expression: <DEFAULT_PARTITION>

Application Metrics

Total Resource Preempted: <memory:0, vCores:0>

Total Number of Non-AM Containers Preempted: 0

Total Number of AM Containers Preempted: 0

Resource Preempted from Current Attempt: <memory:0, vCores:0>

Number of Non-AM Containers Preempted from Current Attempt: 0

Aggregate Resource Allocation: 50609 MB-seconds, 27 vcore-seconds, 0 yarn.io/gpu-seconds

Aggregate Preempted Resource Allocation: 0 MB-seconds, 0 vcore-seconds

Show 20 entries

Search:

Attempt ID	Started	Node	Logs	Nodes blacklisted by the app	Nodes blacklisted by the system
appattempt_1774860907544_2848_000001	Thu May 14 19:43:37 +0800	http://slave216:8042	Logs	0	0

4.4 输出结果

HDFS 输出路径：`/user/231220069a/task3_output`

部分结果截图：

```
100:1
101:13
105:8
106:1
108:1
121:2
122:1
123:3
125:7
140:6
141:42
144:6
153:25
154:9
155:2
```

取消 下载

五、附件说明

压缩包 231220069_杨力泉_BG_4.zip 包含：

- 源代码：Task1.java、Task2FromTask1.java、Task3.java
- JAR 包：Task1.jar、Task2FromTask1.jar、Task3.jar
- 实验报告：231220069_杨力泉_BG_4.pdf
- 运行方式：README.md